Keith Flesher

(719) 304-4286 | FlesherKM@gmail.com | LinkedIn | keithflesher.github.io/Portfolio

EDUCATION

Montana State University
Bachelor of Science in Mechanical Engineering Technology, minor in materials and aerospace. Bozeman, MT

WORK EXPERIENCE

Irrigation & Equipment Technician **may, 2023 – Present**
Tri-Lakes Irrigation Monument, CO

- Installed irrigation systems, sprinkler heads, manifolds, water features, and related components while performing excavation, landscaping, and site preparation
- Performed equipment maintenance and repairs on small-engine equipment, including trenchers and other landscaping machinery, to keep equipment operational
- Rebuilt a 15 hp Kohler engine used in a skid steer, disassembling, inspecting, repairing, reassembling, and testing the engine for proper operation

CAD & Robotics Development Instructor **Summer 2023 (142 hrs) & Summer 2024 (60+ hrs)**
Monumental Impact **|** Community Maker space Monument, CO

- Repaired, assembled, and maintained 3D printers used for prototyping and hands-on STEM projects
- Taught CAD and robotics fundamentals, including 3D modeling and design using Onshape, to middle school students in engineering and robotics programs
- Supported the design and fabrication of BattleBots projects and community STEM events, assisting with prototyping, assembly, and troubleshooting
- Engaged with local engineering firms to gain exposure to professional design, manufacturing, and engineering workflows, including industry practices for product development

ENGINEERING & TECHNICAL EXPERIENCE

FRC Robotics — Mechanical & Fabrication Lead **May 2023 – July 2026**
BearBotics 4068

- Led the Fabraction operations for 3 years, training and managing a total of 10 students in the operation and maintenance of a 3-axis CNC, laser cutter, lathe, and other machining equipment; managing materials and PDR of the current robot
- Supported the development of competition robots through 6-week build seasons, contributing to prototyping, mechanical design, fabrication, assembly, wiring, and final integration; prototyped, manufactured, and assembled 5 competition robots.
- Developed extensive technical documentation in the operation, CAM, and maintenance of a custom 3-axis CNC machine
- Competed in 12 events across the United States, winning a regional event in Oklahoma and qualifying for the 2025 FIRST Championship (Worlds), competing against 600 teams from around the world

FIRST LEGO League (FLL) — Mentor **August 2024–February 2025**

- Mentored 3 teams of 10–12 students in mechanical design and Python programming, guiding the development and testing of LEGO robots for FIRST LEGO League competitions

PROJECTS

Torque drive socket adapter **jan 2026 – May 2026**

- Developed a customer-focused engineering product through the PLTW engineering design process, from identifying a problem and customer needs through design, prototyping, testing, and final evaluation
- Conducted interviews and gathered feedback from potential users and stakeholders, using their needs and requirements to guide design decisions and product development
- Rapidly prototyped and iterated designs using 3D printing, testing functionality, and refining the product through multiple design iterations for varied environments and applications

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- Evaluated the cost, feasibility, and potential value of the product, developing justification for the proposed design and its implementation
- Presented the project to Colorado Springs titans of industry, demonstrating the program's value and advocating for continued investment in advanced engineering education
- Currently exploring patent opportunities for the developed product

Kohler 23 engine rebuild **june 2025 – July 2026**

- Diagnosed an engine failure in a 15 hp Kohler skid-steer engine, identifying the problem before completely disassembling the engine for inspection
- Disassembled the engine to the pistons and cylinder assembly, studying the function and purpose of each component throughout the teardown and repair process
- Replaced both pistons, re-honed the cylinder walls, cleaned and inspected components, and rebuilt the engine following manufacturer service documentation and specifications
- Recorded component locations, assembly order, and torque specifications and patterns during disassembly to support accurate reassembly
- Reinstalled and rewired the engine, returning the skid steer to mechanical operation

Eagle Project **jan 2022–march 2023**

- Planned and managed a community fence-construction project for a local wildlife area, coordinating with the town, community committees, and local officials
- Wrote and presented a formal project proposal and budget to a board for approval, making cost and material decisions based on project requirements
- Participated in town hall meetings and committee discussions, communicating project goals, addressing concerns, and coordinating logistics
- Researched the wildlife area and planned construction methods to protect wildlife and minimize environmental impact
- Directed a crew of 16 Scouts during the build day, assigning tasks and coordinating materials and work areas while resolving unexpected issues, including arranging replacement materials for damaged components

Certifications & Awards

Eagle Scout | 2023

- Earned Eagle Scout rank and completed 47 merit badges in STEM, leadership, and outdoor skills
- Served as Quartermaster, Patrol Leader, and Senior Patrol Leader, managing equipment and leading 20+ Scouts in meetings, trips, and service projects
- Completed 129+ hours of community service and National Youth Leadership Training (NYLT)

Pikes Peak Regional Industry Capstone Recognition | 2026

- Presented capstone work to the Colorado Springs titans of industry, demonstrating the program's value and advocating for continued investment in advanced engineering education

Adobe Illustrator Certified Professional | 2023

- Completed formal coursework in Adobe Illustrator covering digital illustration, graphic design, and vector-based design techniques

Fusion 360 Certified Professional | Pending
Six Sigma White Belt | Pending

TECHNICAL SKILLS

CAD & Design | Fusion 360, Onshape, SolidWorks, Adobe Illustrator
Manufacturing | 3-axis CNC Machining, FDM Printing, Manual Machining,
Engineering | Mechanical Design, Prototyping, Technical Drawings, GD&T, Design for Manufacturing
Equipment & Tools | CNC Mills, CNC Lathes, 3D Printers, Power Tools, Hand Tools